

Michelle Herrera-Cuen

Computer Engineering Student | Embedded Systems | Robotics & Control | Firmware Development

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Technical Skills: C/C++, Python, Verilog, Microcontrollers, Robotics Simulation, Git, Linux, Altium Designer

Education

Bachelor of Science in Computer Engineering

California State University, Long Beach | Expected 2027

Experience

Honors Thesis: Configurable Robotics Firmware Platform

Undergraduate Research Assistant Long Beach, CA | Jan 2025 - Present

Designed a lightweight robotics ecosystem inspired by PX4/Pixhawk enabling no-code/low-code deployment for ground, marine, and articulated robots. Built a desktop GUI to define robot topology, sensors, and kinematic constraints; auto-generated and flashed firmware to ESP32, Teensy, Arduino, and STM32 MCUs. Architected modular abstraction layers for sensor integration, communication (I2C, SPI, UART), telemetry, PID tuning, and calibration

Bio-Inspired Robotic Fish Research | CSULB PACK Lab

Undergraduate Research Assistant Long Beach, CA | Jan 2025 - Present

Modeled fish-like underwater propulsion using Python and embedded C, analyzing tail curvature, frequency, stiffness, and hydrodynamic forces. Developed numerical simulations and control algorithms for energy-efficient swimming and autonomous stability.

Cybersecurity Intern

Blizzard Entertainment — Los Angeles, CA | Summer 2024

Supported cybersecurity operations within a large-scale production environment at a AAA game studio. Gained exposure to secure systems, incident response processes, and cross-functional engineering collaboration in a fast-paced production setting.

Jack in the Box

Team Lead

Torrance, CA | June 2022 - Present

Led shift operations, training, and quality control in a high-volume environment; developed leadership, reliability, and communication skills.

Projects

Embedded Space Invaders Game

Developed an embedded game in C on a TM4C LaunchPad. Integrated Nokia 5110 LCD graphics, ADC-based potentiometer input, GPIO interrupts, and PWM-driven DAC audio. Implemented real-time firmware architecture with collision detection and interrupt-based timing.

Autonomous Robotics Systems

Designed object-following and wall-following robots using IR sensors, DC motors, and microcontroller-based control. Implemented UART, I2C, SPI, ADC, PWM, and interrupt-driven navigation logic.

Associations

Women in Computing (WIC)

President | Aug 2024 - Present

Lead technical workshops, hackathons, and mentorship initiatives. Led development of a Python-based educational app teaching marine science concepts during a 2025 hackathon.

Society of Automotive Engineers

Data Acquisition Team Lead | Sept 2024 - Present

Led ESP32-based data acquisition system development. Implemented LoRa communication between ESP32 and Raspberry Pi for real-time sensor data collection.

